

ACG Performance Benchmarks & Merchant Onboarding

Empirical Latency Measurements & Integration Velocity

1. Canonical Cold-Run Benchmark

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ACG EMPIRICAL BENCHMARK: TIME-TO-FIRST-AI-TRANSACTION
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Execution Milestones (Cold-Start In-Memory Test):
+██ 1. Gateway Boot & Policy Engine: 250.30 ms
+██ 2. Catalog Ingestion & Truth Link: 0.41 ms
+██ 3. Ed25519 Principal Mandate Sign: 3.15 ms
+██ 4. 6-Phase Zero-Trust Agent Checkout: 49.95 ms

■ TOTAL TIME-TO-FIRST-AI-TRANSACTION (Cold Run): 303.81 ms
+██ Gateway Response Status: 201 Created
+██ Razorpay Order Created: order_d753597e364a8240
+██ Policy Version Pinned: pol_v1.0.0
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```

Benchmark Metadata:

- * **Canonical Headline Value:** 303.81 ms (Measured cold-run end-to-end order creation)
- * **Command:** npm run benchmark
- * **Test Environment:** Node.js v22 (x64 Windows 11), in-memory SQLite ACID store.

2. Merchant Onboarding Velocity (10–12 Minutes)

Setup Phase	Description	Estimated Time
Step 1: Environment & Keys	Configure .env with Razorpay API credentials	~3–5 mins
Step 2: Policy DSL Definition	Configure JSON Policy DSL (Allowed categories & transaction caps)	~2 mins
Step 3: Catalog Database Link	Connect SQLite/Postgres product catalog	~5 mins
Total Integration Velocity	Complete Merchant Setup	~10–12 minutes

3. Appendix: Historical Benchmark Runs (Reference Only)

- * 2026-09-01 Initial Prototype: 338.08 ms
- * 2026-09-02 Test Run B: 295.98 ms
- * 2026-09-03 Pre-Freeze Run: 282.94 ms
- * **Active Verified Canonical Baseline:** 303.81 ms